In [1]:
```
import pandas as pd

data = {
    'User ID': [1,2,3,4,5,6,7,8,9,10],
    'Gender': ['Male','Male','Female','Female','Male','Female','Male','Female','Mal
    'Age': [19,35,26,27,45,46,30,40,50,28],
    'EstimatedSalary': [19000,20000,43000,57000,80000,90000,60000,75000,100000,5000
    'Purchased': [0,0,0,0,1,1,1,1,1,0]
}

df = pd.DataFrame(data)
df.to_csv('Social_Network_Ads.csv', index=False)
```

In [2]:
```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
df
```

Out[2]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	1	Male	19	19000	0
1	2	Male	35	20000	0
2	3	Female	26	43000	0
3	4	Female	27	57000	0
4	5	Male	45	80000	1
5	6	Female	46	90000	1
6	7	Male	30	60000	1
7	8	Female	40	75000	1
8	9	Male	50	100000	1
9	10	Female	28	50000	0

In [3]:
```
df.head()
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	1	Male	19	19000	0
1	2	Male	35	20000	0
2	3	Female	26	43000	0
3	4	Female	27	57000	0
4	5	Male	45	80000	1

In [4]:
```
features=df.iloc[:,[2,3]].values
```

```
label=df.iloc[:,4].values
features
```

Out[4]:
```
array([[    19,  19000],
       [    35,  20000],
       [    26,  43000],
       [    27,  57000],
       [    45,  80000],
       [    46,  90000],
       [    30,  60000],
       [    40,  75000],
       [    50, 100000],
       [    28,  50000]])
```

In [5]:
```
label
```

Out[5]:
```
array([0, 0, 0, 0, 1, 1, 1, 1, 1, 0])
```

In [6]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

In [7]:
```
for i in range(1,401):
    x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,ran
    model=LogisticRegression()
    model.fit(x_train,y_train)
    train_score=model.score(x_train,y_train)
    test_score=model.score(x_test,y_test)
    if test_score>train_score:
        print("Test {} Train{} Random State {}".format(test_score,train_score,i))
```

In [8]:
```
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_
```

In [9]:
```
finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)
```

Out[9]:
▾ LogisticRegression ⓘ ?

▸ Parameters

▸ Fitted attributes

In [10]:
```
print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))
```

```
1.0
1.0
```

In [11]:
```
from sklearn.metrics import classification_report
```

In [12]:
```
print(classification_report(label,finalModel.predict(features)))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	5
1	1.00	1.00	1.00	5
accuracy			1.00	10
macro avg	1.00	1.00	1.00	10
weighted avg	1.00	1.00	1.00	10

In [ ]: